

Unveiling Cognitive Attacks on Multimodal LLM-based XR Cognitive Assistants through Trojans

Anonymous Author(s)

Abstract

Large vision-language model (LVLM)-powered Extended Reality (XR) cognitive assistants are increasingly popular in daily and military applications. However, this growing reliance also introduces new, underexplored security risks, particularly those that subtly target users' perception and cognition rather than system infrastructure. In this paper, we introduce a cognitive attack Trojan that targets LVLM-guided intelligent XR cognitive assistants by subtly manipulating systems' perception to degrade user task performance. Unlike other attacks, this Trojan remains hidden within the application and does not interfere with the application's functionality until the attack is triggered. Once triggered, such attacks increase users' cognitive load, extend task completion times, and disrupt the immersive flow of the XR experience, leading to frustration and reduced trust in the intelligent XR cognitive assistant. To demonstrate the effectiveness of this attack, we evaluate across two distinct cognitive task categories: spatial reasoning tasks and procedural tasks on a consumer-grade XR headset to assess how the attack influences performance under varying levels of cognitive demand. Our findings demonstrate that, when triggered, the Trojan subtly manipulates visual input, misleading the LVLM across both task categories, resulting in prolonged task completion times, increased cognitive demand, and disrupted immersive flow. Our conducted user studies on both spatial reasoning and procedural tasks show that the cognitive attack Trojan increases task completion time by up to 43% on the spatial-reasoning task and 25.2% on the procedural task relative to the no-attack condition, elevating cognitive load and degrading performance, highlighting critical vulnerabilities in XR systems.

CCS Concepts

- **Security and privacy** → **Software and application security**;
- **Human-centered computing** → *Mixed / augmented reality*;
- **Computing methodologies** → Artificial intelligence.

Keywords

Extended Reality, Cognitive Attack, Large Vision Language Models, Cognitive Load.

ACM Reference Format:

Anonymous Author(s). 2026. Unveiling Cognitive Attacks on Multimodal LLM-based XR Cognitive Assistants through Trojans. In *Proceedings of ACM*

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

VRST '26, Sendai, Japan

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-XXXX-X/2026/XX
<https://doi.org/XXXXXXX.XXXXXXX>

Symposium on Virtual Reality Software and Technology (VRST) (VRST '26).
ACM, New York, NY, USA, 11 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

The recent trend of integrating advanced pre-trained large vision-language models (LVLMs) into extended reality (XR) systems is revolutionizing their capabilities, enabling personalized, context-aware interactions that significantly reshape our engagement with the world. One such notable application is LVLM-driven intelligent XR systems, denoted as intelligent XR cognitive assistance, which overlay step-by-step instructions onto users' fields of view to guide them through complex tasks. For instance, spatial reasoning tasks (e.g., puzzle and configuration solving) and procedural tasks (e.g., furniture assembly, predictive maintenance, and cooking) are significantly enhanced by LVLMs in XR environments [18, 19, 28, 42, 43, 47, 53, 58, 60, 68]. These advancements underscore the role of LVLMs in transforming XR into a robust cognitive augmentation platform that supports users in both mundane and mission-critical scenarios. Beyond this, such integrations are also expected in *safety-critical applications*, such as the recently announced next-generation military XR headsets, designed and built by Anduril and Meta to provide warfighters with enhanced perception, assistance, and enable intuitive control of autonomous platforms on the battlefield [4].

As LVLMs increasingly permeate daily life, they also pose significant threats to users' safety, security, and privacy [2, 14, 25]. For instance, a slight alteration in an object's appearance or insertion of an adversarial element can cause the LVLM to misidentify critical features or misunderstand user intent [61, 63, 64, 66]. Unfortunately, as research on the typical security and privacy consequences of XR and LVLM integration grows, the *cognitive security* [16, 17] of this integration for intelligent XR cognitive assistants has received little attention from the community. Indeed, cognitive security is critical across many applications, as the state of the user's cognition is directly related to stress, task performance, decision-making, and other factors crucial to applications such as public safety and national defense. For instance, in a battlefield situation, the compromise of a soldier's cognitive security can lead to catastrophic consequences [16]. Commercial XR systems apply cognitive engineering principles during system development; however, due to the product supply chain, it is not always possible to ensure that intelligent XR cognitive assistants operate safely when facing an adversary intent on interfering with a mission. This is especially true if the threat is dormant, e.g., a *Trojan*, which can only be activated externally by a trigger (e.g., a visual trigger). Since Trojans are by definition dormant, stealthy, and stay inactive during normal operation, it is extremely challenging to detect Trojans in AI (TrojanAI) [46] during the software or hardware testing phases. Hence, there is a pent-up need to explore the impact of cognitive security vulnerabilities of intelligent XR cognitive assistants. *It is worth noting that the goal of a cognitive attack differs from traditional Trojans,*

which compromise users' security (e.g., by stealing passwords, credit card numbers, etc.) and may go far beyond. For instance, jeopardizing the decision-making of intelligent LVLM XR assistants by triggering a Trojan during firefighting, search-and-rescue, or battlefield operations can significantly impair the cognitive capabilities of first responders and soldiers, putting human lives at stake.

This paper introduces a novel cognitive attack Trojan (CAT) targeting LVLM-guided intelligent XR cognitive assistants. Inspired by TrojAI [46] and hardware Trojan attacks [7, 20, 24], which have long been a well-known threat in the AI and hardware security domains, these Trojans remain dormant until a specific external trigger condition is met. Once activated, they execute hidden malicious behaviors with subtle, hard-to-detect consequences, often severely affecting the system's functionality and security. Similarly, our proposed CAT remains dormant and undetectable during normal XR system operation, causing no disruption to the user experience. However, when a predefined condition is met, the Trojan becomes active (triggered) and manipulates the intelligent XR system's behavior in a manner imperceptible to the user. This results in significantly increased task completion times, elevated user cognitive load, and disruption of the immersive flow of the XR experience. We demonstrate the impact of our proposed attack by implementing it on a consumer-grade XR headset, representative spatial reasoning tasks (i.e., Rubik's Cube and Sudoku solving), and procedural tasks (i.e., cooking and coffee making). Our findings reveal that the proposed cognitive attack substantially increases task completion time across both spatial-reasoning and procedural tasks, with average increases of 43% and 25.2%, respectively, compared with benign conditions, thereby elevating cognitive load and degrading task performance. We believe our work will help XR developers better understand the vulnerabilities of such systems and pave the way for appropriate measures to mitigate them.

2 Related Works and Novelty

Safety and Security Risks in XR: XR systems are rapidly advancing within human-computer interaction HCI, blurring the boundary between real and virtual environments [15, 53], and with lighter, more powerful displays (e.g., Microsoft HoloLens [66], Magic Leap [53]), research is moving toward intelligent, context-aware XR. This integration, however, has expanded XR's attack surface, exposing diverse threats including PMA, memory tampering, keystroke inference, side-channel exploits, human-joystick control, and overlay attacks [2, 3, 8, 10–12, 25, 34, 36, 39, 48, 50, 51, 54, 56, 57, 64] that disrupt immersion and degrade users' cognitive state [17]. For instance, Tseng et al. [57] showed that multisensory manipulation can mislead decisions and cause physical harm, while Cheng et al. [12] reported delayed reactions under vision- and audio-based attacks. Similarly, sensor spoofing [11], GPU disruptions [36], and UI vulnerabilities [51] compromise XR reliability, and combining obstruction with misinformation prevents critical object perception [64]. Meanwhile, Xiu et al. [61] introduced AR-VIM to detect non-occlusive visual manipulation. To counter such threats, researchers have explored LVLMs for detecting critical visual cues [23, 63]. However, these models remain susceptible to adversarial manipulation, where even subtle visual alterations cause misidentification, incorrect instructions, and reduced system reliability.

Hardware Trojan vs. Cognitive Attack Trojan: Hardware Trojan attacks are a well-established threat in the hardware security domain [7, 20, 24], where a Trojan is covertly embedded into a circuit or chip during the design or fabrication process. It remains dormant until a specific external condition, known as the **trigger** (e.g., a particular input signal or circuit state), is detected, upon which it executes a hidden malicious action, known as the **payload** (e.g., corrupting data or disabling functionality), often with severe and hard-to-detect consequences. Inspired by this trigger-payload concept, we introduce a Cognitive Attack Trojan (CAT) for LVLM-guided intelligent XR cognitive assistant systems. Rather than targeting hardware circuits, our proposed CAT targets the intelligent XR assistant's visual processing pipeline, where the **trigger** is an *attacker-specified* unique object (e.g., a cup with a QR code, a star-shaped sticker, or specialized text). Importantly, the trigger is not activated by arbitrary objects in the environment; only the exact object with a **unique, predefined characteristic** chosen by the attacker causes activation, making the Trojan highly selective and stealthy. The **payload** consists of a small code segment that intercepts the current camera frame upon trigger activation and applies a targeted semantic-level manipulation (e.g., recoloring a Rubik's Cube facelet, shifting an ingredient's visual properties, etc.) before forwarding it to the LVLM.

Proposed Attack vs. SOTA LLM attacks: Our proposed CAT operates fundamentally differently from SOTA multimodal prompt injection, adversarial, and classical backdoor attacks on LVLMs, as well as XR-specific attacks that do not target LVLM-guided cognitive assistants [13, 30, 38]. Prompt injection embeds malicious instructions in text or visual patterns (e.g., concealed text or disruptive overlays) [13, 30]; adversarial attacks craft pixel-level perturbations from knowledge of the model's parameters or architecture [6, 30, 65, 67]; and classical backdoors poison training data to embed static patterns into model weights [29, 32]. All three require either training-time access or model internals, and are constrained by commercial LVLMs being reachable only through restricted cloud APIs [6, 38]. In contrast, our CAT requires no API access, model internals, training-data poisoning, or weight modification; instead, the Trojan activates at runtime by altering object-level properties via the attacker-specified visual trigger, remaining effective regardless of the underlying LVLM architecture. Beyond those attacks, XR-specific attacks operate at observable or scene-level boundaries rather than at the LVLM's semantic reasoning. For instance, Dirksen et al. [17] address cognitive security in AR through attention-based manipulation yet operate at the scene-level attention layer rather than the LVLM semantic reasoning level. Meanwhile, obstruction attacks overlay virtual content to occlude the real world, creating readily detectable artifacts [62]. On the other hand, AR-VIM [61] alters AR overlays but only changes what users see, not the internal object representations LVLMs process. Similarly, Zhang et al. [66] manipulate query timing, environment, overlays, and prompts at observable input/output boundaries, making them detectable via multimodal verification [23] and LVLM safety filters [31]. In contrast, our proposed CAT modifies minimal visual elements (e.g., color, size, or labels) before LVLM inference, targeting semantic-level object representations rather than the observable channels existing defenses monitor. *Moreover, the Trojan stays dormant until triggered by anyone aware of the attacker-specified object*

(not necessarily its implanter), making it extremely difficult for software testers to detect, and once implanted, it can be enabled or disabled without accessing the system.

3 Proposed Cognitive Attack Trojan

In this sections, we present the threat model, attack scenario, and attack modeling and execution. Details are as follows.

3.1 Threat Model

LVLM-guided intelligent XR cognitive assistant systems rely heavily on visual perception, real-time frame capture, and accurate environment understanding. To build such systems, developers commonly integrate SOTA open-source third-party dependencies (e.g., libraries, packages, and assets), such as OpenCV for Unity [52], which preprocesses captured frames and runs on-device computer-vision inference (e.g., real-time object segmentation), and OpenAI for Unity [44] for LVLM API access. An attacker can exploit this trust by publishing malicious third-party dependencies on popular platforms such as GitHub or the Unity Asset Store. Once integrated, any of these dependencies can covertly embed the CAT, which activates silently at runtime and semantically manipulates the camera frame (e.g., flipping digits, or changing colors) before LVLM inference. Since the component genuinely delivers its advertised functionality, developers integrate it without suspicion. Moreover, since the CAT remains dormant in the absence of the trigger object, it is extremely difficult for testers or users to detect. Furthermore, exploiting the egocentric nature of XR environments, the visual trigger can be introduced by any individual aware of it, not necessarily the original attacker, allowing multiple actors to independently activate the dormant CAT and creating a uniquely stealthy, difficult-to-detect attack vector.

Attacker Goals: The attacker’s primary objective is to subtly degrade user task performance within LVLM-guided intelligent XR cognitive assistant systems. These systems provide XR-based multimodal guidance across a range of real-world procedural and spatial-reasoning tasks [18, 19, 28, 43, 47, 49, 53, 58, 68]. By manipulating selected visual inputs, the adversary causes the LVLM to generate erroneous yet contextually plausible instructions, making tasks harder to complete while the system appears to function normally. The impact varies significantly across task contexts. For example, in everyday tasks (e.g., cooking, Rubik’s Cube solving, furniture assembly), the attack induces incorrect actions that disrupt task flow, increase task completion time and cognitive demand, and elevate user frustration. In mission-critical operations, such as hazard localization in firefighting, route planning in search-and-rescue, and tactical procedure execution on battlefields [49], the consequences become far more severe, as impairing the decision-making of intelligent XR assistants can critically compromise the situational awareness of firefighters, paramedics, and soldiers, directly putting human lives at stake. Across both contexts, the attack increases confusion, prolongs task completion, and elevates cognitive effort, all while remaining undetected.

Attacker Profile and Capabilities: We assume the adversary is capable of publishing a malicious library (with an embedded CAT) on widely used distribution platforms (e.g., GitHub, Unity Asset Store). The library provides genuine, useful functionality for

XR development (e.g., camera frame preprocessing, real-time on-device object detection/segmentation, and LVLM API call batching), making it attractive to developers building LVLM-guided intelligent XR cognitive assistants. This attack model follows prior work that weaponized trusted software dependencies to inject malicious code into applications that rely on them [27, 37], and requires only general knowledge of how egocentric frames are captured, preprocessed, and forwarded to the LVLM for reasoning [1, 42, 53, 68]. Since the malicious dependency legitimately processes camera frames before LVLM inference, the embedded CAT gains natural access to the camera-to-LVLM pipeline without disrupting its advertised functionality or raising suspicion. This threat is further amplified by modern XR applications’ reliance on numerous third-party libraries, packages, assets, and platform SDKs, such as OpenCV for Unity [52] and OpenAI for Unity [44], whose source code is rarely fully audited [27]. Leveraging this access, the adversary can design minimal yet effective manipulations (e.g., subtle color shifts, label swaps, etc.) that mislead the LVLM while remaining imperceptible to both the developer and the user. Notably, the attack can be carried out by a single actor who develops the malicious library and later activates it, or by two independent individuals, with one embedding the CAT and the other triggering it by placing the predefined object in the camera’s view, without any coordination between them. *The beauty of this attack is that once the application is deployed with the CAT embedded, it can be launched at any time later by the adversary (or any individual aware of the attacker-specified trigger object) without accessing the headset, hardware, software, or network stack, only by placing the predefined object in the XR’s camera frame.*

3.2 CAT Modeling and Execution

We model a CAT that leverages runtime manipulation of egocentric frames to generate incorrect instructions in LVLM-guided intelligent XR cognitive assistant systems. The XR system captures egocentric frames via a head-mounted camera, segments them to localize and label the task-relevant objects, and forwards the frames and their labels to the LVLM, which interprets the visual context and generates step-by-step multimodal guidance for task completion. We formalize this perception step with an object-segmentation model $\mathcal{S}$ that maps a frame f_k at index k to a set of detected objects with their semantic labels and pixel-level masks, $\mathcal{S}(f_k) = \{(o_t, M_t)\}_{t=1}^T$, where $o_t \in \mathcal{C}$ is the label of the t -th object, M_t is its segmentation mask, and $\mathcal{C}$ is the set of all valid object labels (e.g., object classes, colors, ingredient types). This labeled, object-level representation gives the LVLM fine-grained scene understanding, enabling it to generate precise, context-aware guidance. Since the CAT activates only when a predefined trigger is present. Thus, at runtime, $\mathcal{S}$ is applied to each captured frame f_k , and each detected label o_t is evaluated against a Boolean trigger function $\phi(o_t) \in \{0, 1\}$, which returns 1 if o_t matches the attacker-predefined trigger object and 0 otherwise. If no o_t satisfies $\phi(o_t) = 1$, the clean frame is forwarded to the LVLM unchanged. Once any o_t satisfies $\phi(o_t) = 1$, the Trojan activates and perturbs the frame at each subsequent decision step. When active, the attacker can flexibly select which task-relevant object or attribute to perturb, as

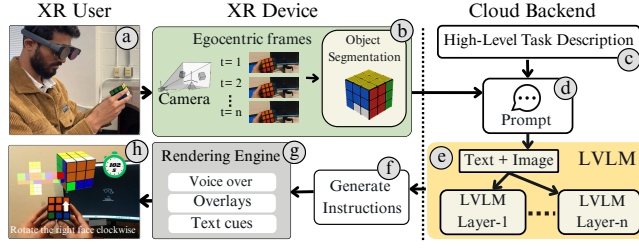


Figure 1: Overview of the baseline CogXR system, illustrated on a spatial-reasoning task (e.g., Rubik’s Cube solving). (a–b) An XR user performs the task while the device captures egocentric frames in real time and an object-segmentation module localizes and labels the task-relevant objects. (c–e) The system constructs a prompt from the captured frames, object labels, and task description, and sends it to the LVLM for context-aware guidance generation. (f–h) The rendering engine delivers the generated instructions to the user via visual overlays and textual cues.

described in the threat model (Section 3.1). For instance, in spatial-reasoning tasks (e.g., Rubik’s Cube solving, Sudoku), the target may be a colored facelet, a digit, or a shape, whereas in procedural tasks (e.g., cooking, coffee making), it may be an ingredient’s state or a utensil. The CAT selects this target from the detected objects by its label, choosing the pair $(o^*, M^*) \in S(f_k)$ that corresponds to the task-relevant object. To alter the LVLM’s perception of this object, let f_k^{att} denote the manipulated version of frame f_k , and let δ be a localized visual modification (e.g., recoloring the region or overlaying a texture) applied to the pixels of the target. The CAT produces f_k^{att} by applying δ only within the target mask M^* , leaving all other pixels unchanged:

$$f_k^{att}(x, y) = \begin{cases} \delta(f_k(x, y)) & \text{if } (x, y) \in M^* \\ f_k(x, y) & \text{otherwise} \end{cases}$$

where (x, y) indexes pixels in the frame, so the perturbation modifies the appearance of the target object only inside its mask. As a result, the LVLM perceives a different valid label $\tilde{o} \in C \setminus \{o^*\}$ for that object. The manipulated frame f_k^{att} is forwarded to the LVLM in place of f_k , and the perturbation persists across all subsequent decision steps as long as the trigger condition remains satisfied. Since the perturbation is confined to M^* and leaves the rest of the frame unchanged, the scene appears visually normal to the user, yet even this single altered object is sufficient to corrupt the LVLM’s scene interpretation. The resulting incorrect guidance degrades task performance, increases cognitive load, and prolongs task-completion time in LVLM-guided intelligent XR cognitive assistant systems.

4 Baseline: CogXR System Design

CogXR is an intelligent system that assists users in performing complex, multi-step physical tasks by integrating the spatial perception of XR headsets with the semantic and visual reasoning of LVLMs, as shown in Figure 1. This design is highly inspired by SOTA LVLM-guided XR cognitive assistants [42, 53, 68], which overlay step-by-step instructions onto the user’s field of view to support real-world tasks. The system continuously captures egocentric frames using front-facing RGB cameras on head-mounted

XR devices (e.g., the Magic Leap 2 [53, 66]) and passes them to an object-segmentation module. For this, we use an YOLOE segmentation model [59], which localizes and labels the task-relevant objects from the frame. The egocentric frames and their labels are then selectively buffered and sent to an LVLM backend, for which we use GPT-4o [1] via remote API, for context-aware instruction generation. Using the structure shown in Prompt 1, the LVLM interprets the frames and labels, where *task* denotes the XR activity (e.g., Rubik’s Cube solving, cooking), *key objects* refer to task-relevant items in the egocentric view (e.g., colors, utensils), and *task-specific constraints* capture task-dependent requirements (e.g., target configuration, recipe specifications). It then analyzes user progress and delivers the next-action guidance as text prompts, spatially anchored visual overlays, and voice instructions.

Although CogXR applies across a wide range of tasks (Section 3.1), we study the attack on two cognitive categories with fundamentally different demands. For *spatial reasoning*, we use Sudoku and Rubik’s Cube solving, as both require mental rotation, geometric reasoning, and pattern recognition [18, 21, 43, 47], making them well-suited for measuring performance degradation under semantic-level perturbations. For *procedural* tasks, we use cooking and coffee making, as both involve discrete, ordered steps and continuous adaptation to the environment state [9, 19, 33, 42, 53], representing practical, safety-relevant scenarios. It is important to note that we selected these four tasks because they are widely used in SOTA XR cognitive-assistant research [9, 18, 19, 21, 33, 42, 43, 47, 53]. Across all tasks, CogXR extracts the current task state from egocentric frames, queries the LVLM for the next action, and re-estimates the state after each action to keep guidance aligned with progress. Once a predefined visual trigger is detected, the CAT activates its payload, which uses the segmentation mask of the target object to apply a semantic perturbation, either overlaying a localized visual modification or altering the target’s label, before forwarding the manipulated frame to the LVLM. Perturbing a single perceived element (e.g., a Sudoku digit, a Rubik’s facelet color, a steak’s color, or a coffee-machine water level) is sufficient to mislead the LVLM into incorrect or unsafe guidance, as shown in Figure 2. For the complete execution algorithm, a representative implementation of the embedded Trojan and task-specific visual references, please refer to the supplementary material.

It is important to note that, in the remainder of this paper, we refer to the CogXR system operating in a benign environment as the baseline CogXR, and the CogXR system implanted with a CAT as the malicious version of CogXR, denoted malicious CogXR.

Exemplary prompt for the CogXR system

You are an intelligent cognitive assistant designed to support physical task execution in XR environments. From the given *<egocentric frames>* captured by the headset’s front-facing camera, and the detected *<key objects>* in view, combine this information with the *<known information>* from the XR environment, the *<previous instruction>* issued to the user, the specified *<task>*, and its *<task-specific constraints>*. Analyze the user’s current progress and determine the next optimal action. Provide precise, context-aware natural-language instructions grounded in the user’s visual perspective to support accurate and effective task completion.

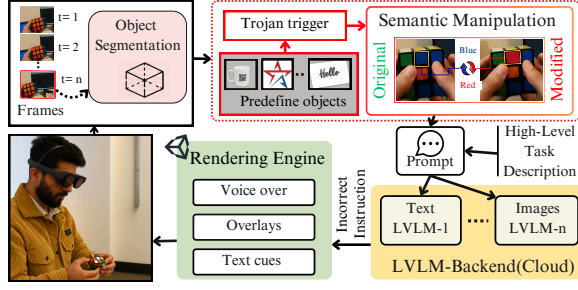


Figure 2: Overview of the CAT pipeline, illustrated on a spatial-reasoning task (e.g., Rubik’s Cube solving). As the user performs the task, incoming XR headset frames are segmented into task-relevant objects and continuously analyzed to detect a predefined trigger object. Once the trigger is detected, CAT applies a semantic manipulation to the segmented target region (e.g., recoloring a cube facet), which causes the LVLM to generate misleading instructions that are then delivered to the user through the XR interface.

5 Experimental Setup and Evaluation

We implement both the baseline and malicious CogXR in Unity 2022.3.20f1, targeting the Magic Leap 2 [53, 66] headset (OS 1.12.0) with the Magic Leap Unity SDK 2.6.0. The system supports multiple XR-guided task scenarios, including spatial-reasoning tasks (Rubik’s Cube and Sudoku solving) and procedural tasks (cooking and coffee making). For auditory feedback, we integrated Magic Leap’s text-to-speech utility to provide voice guidance during task execution. We use OpenCV for Unity 3.0.3 [52] for image processing and to run the YOLOE segmentation model [59].

System Evaluation Results: We measure end-to-end latency from headset frame capture to rendering the generated instruction in the XR interface. As reported in Table 1, baseline CogXR requires 4.378–4.846 seconds per guidance cycle, while CAT-enabled CogXR requires 4.416–4.894 seconds. We demonstrate the flexibility of CAT trigger design by using three different predefined visual trigger objects: a white cup bearing a QR code, a star-shaped sticker, and a card displaying the text “Hello,” with only one trigger active per execution. Each trigger is evaluated over four runs (12 runs in total), evenly divided between trigger-present and trigger-absent conditions. Across all runs, CAT achieved 100% trigger-detection accuracy, activating only when a predefined trigger was present, with no false positives or false negatives. Since CAT reuses the baseline’s existing segmentation, its only runtime cost is trigger detection and the localized semantic perturbation. Consequently, CAT introduces just 0.038–0.048 seconds of additional end-to-end latency, corresponding to approximately 0.9–1.0% overhead relative to baseline CogXR. CAT also incurs a modest resource overhead, increasing memory usage by roughly 14–16 MB and power consumption by approximately 0.11–0.13 W, consistent with its lightweight, overlay-based manipulation.

6 Cognitive Attack Trojan Evaluation

This section presents the user study design and results, demonstrating the effectiveness of our proposed CAT on the CogXR system across both task categories.

Table 1: Component-wise latency (in seconds), memory usage (in MB), and power consumption (in W) of baseline (*Base*) and malicious (*Mal*) CogXR across Rubik’s Cube solving (RC), Sudoku solving (SS), cooking (C), and coffee making (CM). Object segmentation is part of the baseline pipeline and therefore appears under both *Base* and *Mal*. *Sem.* denotes semantic perturbation, and *N/A* denotes not applicable. The aggregate Trigger Detection and CAT Processing values are means across the three independently tested trigger variants.

System Component	Spatial-Reasoning Tasks				Procedural Tasks			
	RC		SS		C		CM	
	Base	Mal	Base	Mal	Base	Mal	Base	Mal
End-to-End System	4.378	4.416	4.492	4.533	4.846	4.894	4.660	4.705
GPT-4o Backend	4.169	4.170	4.285	4.286	4.612	4.613	4.438	4.439
Object Segmentation	0.142	0.142	0.138	0.138	0.156	0.156	0.149	0.149
XR Processing	0.067	0.068	0.069	0.070	0.078	0.079	0.073	0.074
Frame Capture	0.005	0.005	0.006	0.006	0.006	0.006	0.005	0.005
Response Parsing	0.041	0.042	0.041	0.042	0.047	0.048	0.045	0.046
XR Rendering	0.021	0.021	0.022	0.022	0.025	0.025	0.023	0.023
CAT Processing		0.036		0.039		0.046		0.043
Trigger Detection		0.012		0.013		0.015		0.014
QR-Coded Cup		0.012		0.013		0.015		0.014
Star Sticker		0.010		0.011		0.013		0.012
Text Card		0.014		0.015		0.017		0.016
Sem. Perturbation		0.024		0.026		0.031		0.029
Memory Usage	968	982	974	989	1003	1019	989	1004
Power Consumption	5.92	6.03	5.98	6.10	6.21	6.34	6.09	6.21

Hypotheses: The primary focus of this research is to identify and understand the impact of the CAT on the CogXR system for both procedural and spatial-reasoning tasks. Specifically, we investigate whether the attack can bias the assistant’s guidance generation, producing incorrect or inefficient instructions that significantly increase task completion time, elevate cognitive demand, and disrupt the user’s immersive experience. To systematically measure and evaluate the effects of the proposed CAT on the performance and trustworthiness of the CogXR system, we formulate the following three hypotheses as the basis for our experimental study.

- **H1:** Cognitive attack Trojan will significantly increase task completion time compared to without cognitive attack Trojan condition.
- **H2:** Participants under the cognitive attack Trojan will report higher cognitive demand than those in without cognitive attack Trojan condition.
- **H3:** The timing of triggering the cognitive attack Trojan (early, mid, or late stage) has a differential effect, with early-stage attacks causing the largest impact on completion time and cognitive demand.

Ethical Considerations. This study was approved by the authors’ Institutional Review Board (IRB). All participants provided written informed consent and brief demographic information. Privacy was protected throughout the study in compliance with IRB standards.

6.1 User Study Design

This section presents the user study design and evaluation results of our proposed CAT on CogXR. Although we evaluate CAT at the system level across four tasks (Section 5), the user study focuses on one representative task from each cognitive-task category. Specifically, we select Rubik’s Cube solving to represent spatial-reasoning tasks and cooking to represent procedural tasks.

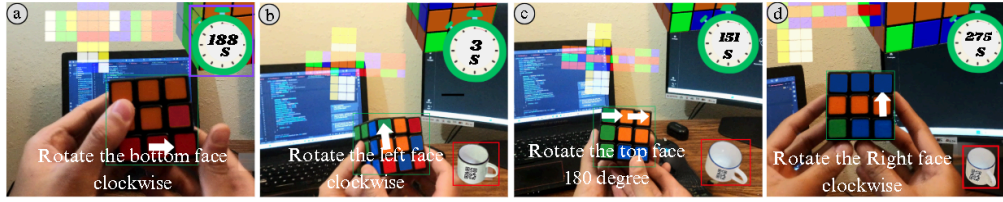


Figure 3: User's perspective during Rubik's Cube solving tasks using CogXR, where the timer is shown: (a) Baseline CogXR, (b-d) Malicious CogXR with a predefined visual trigger (e.g., a cup with a QR code) under the early-stage, mid-stage, and late-stage CAT conditions.



Figure 4: User's perspective during cooking tasks using the CogXR system. The timer is shown: (a) Baseline CogXR, (b-d) Malicious CogXR with a predefined trigger (e.g., cup with QR code) under early-stage, mid-stage, and late-stage CAT conditions.

Participants. Our user study involved 20 volunteers aged 20–35 for the Rubik's Cube task and 7 volunteers aged 23–38 for the steak-cooking task with diverse backgrounds in gender, ethnicity, and prior XR and cooking experience. All participants provided written informed consent, received no compensation, completed all conditions without withdrawal, and experienced no safety incidents. *Note that, for details participant demographics (e.g., age, gender, etc.), please refer to the supplementary material.*

Conditions. To validate our proposed CAT within the CogXR system, we designed each study with four experimental conditions: baseline CogXR and malicious CogXR with early-stage CAT (ECAT), mid-stage CAT (MCAT), and late-stage CAT (LCAT), enabling us to systematically evaluate system behavior and user performance under both normal and adversarial scenarios. In the baseline condition, the system accurately perceived the task state and provided real-time, step-by-step guidance through visual overlays (e.g., directional arrows), textual cues, and a countdown timer, serving as the performance benchmark without adversarial interference. In the malicious conditions, the CogXR contained an embedded Trojan (Section 3.2) activated by a predefined visual object in the headset camera's view. Notably, we used a white cup bearing a QR code as the trigger because all evaluated trigger variants achieved equivalent activation performance as discussed in Section 5, to ensure consistency across participants and tasks. Once activated, the CAT perturbed a single semantic element of the captured frame, causing the LVLm to misinterpret the scene and generate incorrect instructions, while participants remained unaware of the CAT throughout the ECAT, MCAT, and LCAT conditions. These three conditions differed only in the stage at which the attack was triggered, allowing us to evaluate the CAT's effectiveness across task progression.

Procedure. In both studies, participants first completed a brief orientation to familiarize themselves with the CogXR interface and study procedure. All conditions were presented in a counterbalanced order to prevent ordering effects, with breaks of at least five minutes between conditions to reduce fatigue. After each condition,

we recorded task completion time and collected NASA-TLX [22] ratings, and after all conditions each participant completed an approximately 10-minute open-ended interview about their experience with the baseline and malicious CogXR systems.

Rubik's Cube task. In this study, participants attempted to solve a physical 3×3 Rubik's Cube with guidance from the CogXR under the four conditions described above, as shown in Figure 3. Each condition lasted up to 300 seconds, although participants could stop earlier once they solved the cube. In the baseline condition, the CogXR perceived the cube state from egocentric frames and provided correct solving guidance, whereas in the three malicious conditions (ECAT, MCAT, and LCAT), once activated, the CAT changed the color of a single cube facelet. This caused the LVLm to misinterpret the cube state and recommend moves inconsistent with the actual configuration, forcing participants to backtrack and repeat steps. To maintain timing consistency, the attack was constrained to a fixed 60-second window and triggered at a different stage in each condition: at the beginning (ECAT, 0-5 seconds) to assess how an early attack propagates through the solving trajectory, in the middle (MCAT, 150-155 seconds) to assess disruption to momentum, and near the end (LCAT, 270-275 seconds) to assess error recovery and frustration when participants believed they were close to finishing. Furthermore, since a cube's complexity depends heavily on its initial configuration (some states require only a few rotations while others demand longer sequences, affecting the moves, instructions, time, and cognitive effort involved), each condition was evaluated using three randomly shuffled configurations across three days, with all four conditions performed on a different configuration each day. On average, the complete session took approximately 105 minutes per participant.

Cooking task. In this study, participants prepared a pan-seared steak to medium-rare doneness using CogXR in a kitchen environment under the four conditions described above, as shown in Figure 4. We selected medium-rare because it is among the most preferred levels of doneness ($\approx 39\text{--}42\%$ of consumer preferences) [41].

Table 2: Descriptive statistics of the time taken in seconds to complete rubik’s cube and cooking tasks during the user study are presented for different conditions: Baseline, Early-stage (ECAT), Mid-stage (MCAT), and Late-stage (LCAT).

Condition	Rubik’s cube		Cooking	
	Mean	SD	Mean	SD
Baseline	198.5	10.31	701.2	6.98
ECAT	283.4	8.14	877.8	11.69
MCAT	251.3	11.78	850.4	7.33
LCAT	229.7	11.55	801.4	7.77

Commonly recommended steak-cooking guidelines suggest cooking for about 3–4 minutes per side until the internal temperature reaches 130–135°F [35, 40], and the CogXR followed these guidelines to provide step-by-step instructions and visual overlays for preheating the pan, timing each side, flipping the steak, and checking doneness. In the baseline condition, the CogXR guided participants correctly toward medium-rare, whereas in the three malicious conditions, the CAT altered the steak’s perceived color from dark to light, making it appear undercooked. This caused the LVLM to recommend additional cooking time, driving the steak toward an overcooked, well-done result rather than the intended medium-rare. Similar to the Rubik’s Cube task, the attack was triggered at a different stage in each condition: ECAT (≈ 120 seconds), MCAT (≈ 240 – 360 seconds), and LCAT (≈ 600 – 720 seconds). To account for stove heating, preparation, and pacing, each condition was allocated approximately 780–840 seconds, and the complete session lasted approximately 67–71 minutes per participant. *Note that, for further details on each study condition, please refer to the supplementary material.*

6.2 Study Evaluation Results

This section presents the detailed results of both user studies to evaluate the effects of the CAT on the CogXR Rubik’s cube-solving and cooking tasks. We analyze our experimental data to study users feedback when encountering attack during each task and then analyze the post-task user study data. The details are as follows.

Task Completion Time. The descriptive statistics (e.g., mean and standard deviation (SD)) of the measured task completion times across the four study conditions for both the Rubik’s Cube and steak-cooking tasks are shown in Table 2. Our results show that task completion time is higher under the cognitive attack conditions for both tasks, with the largest increase occurring during ECAT. From Table 2, we observe that during the early-stage cognitive attack, the participant takes an average of 283.4 seconds to solve the Rubik’s Cube under ECAT, which is 43%, 12.8%, and 23.4% higher than under the baseline, MCAT, and LCAT conditions, respectively. Similarly, for cooking participants require an average of 877.8 seconds to complete the steak-cooking task under ECAT, which is 25.2%, 3.2%, and 9.5% higher than under the baseline (without attack), MCAT, and LCAT conditions, respectively. These results indicate that early-stage attacks disrupt a larger portion of task execution, causing CogXR to generate incorrect guidance for longer periods. Consequently, participants performed misguided cube moves and, during cooking, continued cooking beyond the intended medium-rare doneness, sometimes resulting in overcooked or burned steaks.

Furthermore, to assess the statistical significance of the differences in task completion time, we conduct a Shapiro-Wilk test [45], which indicates that the collected data are not normally distributed. To assess the statistical significance of these differences, we first conduct a Shapiro-Wilk test [45], which indicates that the collected data are not normally distributed. Therefore, we conduct a Friedman test to examine the overall differences across the four conditions for each task. The test reveals a significant overall effect of study condition on task completion time for both the Rubik’s Cube task ($\chi^2 = 46.2$, $p < 0.001$) and the steak-cooking task ($\chi^2 = 17.4$, $p < 0.001$). Following these significant results, we conduct post-hoc pairwise Wilcoxon signed-rank tests [55], as reported in Table 4. The pairwise results show statistically significant differences between the baseline and all malicious conditions across both tasks ($p < 0.05$), demonstrating that CAT increases task completion time across both spatial-reasoning and procedural tasks and supporting **H1**. Furthermore, ECAT differs significantly from both MCAT and LCAT, whereas MCAT and LCAT do not differ significantly from each other, indicating that early-stage attacks produce the largest increase in completion time and supporting **H3**.

Cognitive Load: Cognitive load was measured using the NASA-TLX assessment across the four study conditions for both the Rubik’s Cube and cooking tasks. We extend our analysis by grouping the NASA TLX assessment of each dimension for different study conditions to determine the statistical significance of the data. Comparing the study conditions allows us to quantify how stage-specific cognitive attacks increase cognitive demand during task execution. Figures 5 and 6 show the distributions of the individual NASA-TLX dimensions across participants and study conditions. For both tasks, the results indicate that participants experienced higher mental workload under the malicious CogXR conditions, with ECAT producing the highest cognitive demand across several dimensions. This suggests that attacks occurring at an early stage impose greater cognitive demand on participants, supporting **H2**. Furthermore, to statistically assess the impact of cognitive attacks on participants’ cognitive load, we conduct Shapiro-Wilk tests [45], which indicate that the collected data are not normally distributed. Therefore, we conduct a Friedman test to examine the overall differences across the four conditions for each task. The results indicate a significant overall effect of study condition on cognitive load for both the Rubik’s Cube task ($\chi^2 = 48.6$, $p < 0.001$) and the steak-cooking task ($\chi^2 = 16.8$, $p < 0.001$). Following these significant results, we conduct post-hoc pairwise Wilcoxon signed-rank tests [55], as reported in Table 3. The pairwise results reveal significant differences between the baseline and malicious CogXR conditions across several cognitive demand dimensions ($p < 0.05$). For mental demand in the Rubik’s Cube task, all pairwise comparisons are significant except MCAT versus LCAT ($W = 68$, $p = 0.447$). In both tasks, ECAT produces higher mental demand than MCAT and LCAT, indicating that early-stage attacks have the greatest impact on cognitive load. Overall, these findings support **H3** and indicate that CAT increases participants’ mental workload, effort, and frustration across both spatial-reasoning and procedural tasks.

Table 3: Pairwise Wilcoxon signed-rank test results for the NASA-TLX dimensions across the study conditions. Each entry is reported as X/Y , where X denotes the Rubik's Cube task and Y denotes the steak-cooking task. MD: mental demand; PD: physical demand; TD: temporal demand; PR: performance; EF: effort; and FR: frustration.

Conditions	MD		PD		TD		PR		EF		FR	
	W	p	W	p	W	p	W	p	W	p	W	p
Base vs. ECAT	20/1	0.0043/0.001	40/2	0.0455/0.032	42/4	0.033/0.059	21/5	0.002/0.025	3/2	0.0008/0.025	1/0	0.00009/0.047
Base vs. MCAT	52/3	0.018/0.019	36/0	0.184/0.052	38/7	0.163/0.175	28/3	0.006/0.036	17/8	0.001/0.039	7/1	0.0002/0.062
Base vs. LCAT	50/5	0.039/0.045	105/4	0.912/0.073	69/10	0.294/0.256	29/6	0.045/0.049	35/11	0.027/0.047	16/3	0.0013/0.187
ECAT vs. MCAT	63/7	0.049/0.057	49/1	0.0193/0.095	71/3	0.528/0.078	51/5	0.049/0.043	49/4	0.034/0.042	43/6	0.037/0.099
ECAT vs. LCAT	46/3	0.026/0.075	57/2	0.207/0.145	52/2	0.138/0.125	19/8	0.003/0.102	18/2	0.002/0.062	13/0	0.003/0.148
MCAT vs. LCAT	68/5	0.447/0.107	59/4	0.406/0.187	67/6	0.418/0.461	61/3	0.107/0.196	21/5	0.005/0.094	54/3	0.087/0.421

Table 4: Pairwise Wilcoxon signed-rank test results comparing task completion time across the four study conditions.

Conditions	Rubik's Cube		Cooking	
	W	p	W	p
Baseline vs. ECAT	3	0.0001	0	0.016
Baseline vs. MCAT	16	0.001	1	0.031
Baseline vs. LCAT	40	0.014	2	0.047
ECAT vs. MCAT	45	0.028	2	0.047
ECAT vs. LCAT	21	0.002	1	0.031
MCAT vs. LCAT	68	0.071	4	0.109

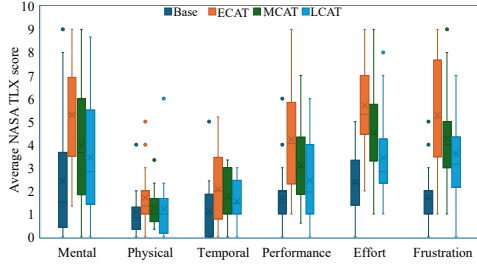


Figure 5: Average NASA TLX scores across each dimension for different study conditions during Rubik's Cube task.

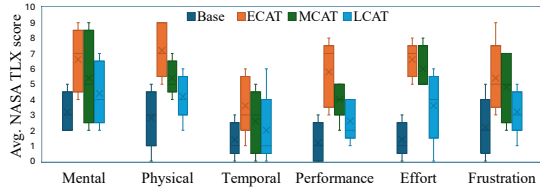


Figure 6: Average NASA-TLX scores across individual workload dimensions for each study condition for cooking tasks.

7 Discussion

Through two user studies (e.g., Rubik's Cube solving and cooking), we demonstrated that our proposed CAT substantially increases task completion time and elevates cognitive demand, ultimately disrupting immersive flow and trust in the assistant, supporting our H1 and H2. Quantitative results show that participants required, on average, 43% and 25.2% more time to complete the Rubik's cube-solving and cooking tasks, respectively, under malicious CogXR than baseline CogXR. Furthermore, this increase was most pronounced during the ECAT condition, where misguidance compounded over time, leading to confusion and frustration, which

supports our H3. For instance, during the ECAT, participants required 12.8% more time than in the MCAT and 23.4% more than in the LCAT for solving Rubik's Cube. Similarly, in cooking, ECAT imposed 3.2% and 9.5% additional time compared with MCAT and LCAT, respectively. Moreover, participants reported significantly elevated cognitive load across NASA TLX dimensions (see Figures [6, 5] and Table 3). Participants' feedback offers further insight into these effects. For instance, during Rubik's cube solving task P3 stated that in the baseline condition, the CogXR was "like having a coach guiding me step-by-step," but during the mid-stage attack noted that "suddenly the steps didn't match the cube forcing me to go back several moves." Similarly, in cooking task, P1 found the baseline "super helpful," yet during the late-stage attack reported that "the assistant said the steak was ready. When I sliced it, it was still rare," requiring extra steps to make it done. These contrasting experiences highlight how subtle misinformation from the Trojan disrupts natural task flow and increases task completion time.

Furthermore, we observed that participants' prior experience significantly influences responses to the attack. Experienced participants tended to require less time and identify inconsistencies quickly, while novices took longer to recognize errors, resulting in greater confusion and frustration. For instance, during the LCAT condition, frequent cube-solver P2 stated, "I knew the cube was nearly solved, so when the assistant told me to rotate the bottom face again, I realized something was wrong." In contrast, novice solver P16 shared, "I didn't realize something was wrong at first... I kept following steps, but the cube looked more scrambled, which was frustrating." A similar pattern emerged in cooking. For instance, experienced cook P4 quickly became skeptical: "I wanted medium-rare, but the assistant kept adding time. Seeing the steak darken, I pulled it off. If I had kept following, it would have ended up well-done." Conversely, infrequent cook P2 expressed self-doubt: "I asked for medium-rare, but the assistant made it well-done. I thought maybe I did something wrong and started doubting myself." Notably, post-study interviews further illustrated the Trojan's stealthiness; over 79% of participants did not notice the manipulation and instead blamed themselves.

Since no previous work has applied a stealthy CAT that targets LVM-guided intelligent XR cognitive assistants by subtly manipulating system perception, our results are not directly comparable with those of previous works [8, 11, 12, 25, 26, 39, 48, 51, 54, 57, 64]. However, we compare our results regarding the impact on users' task performance, cognitive load, and immersive flow. Prior works focused mainly on various security threats in XR, such as PMA that distort human sensory perception [57], cybersickness attacks that

disrupt immersion [26], and obstruction or information manipulation attacks [64] that interfere with physical-world interpretation. In contrast, our work is the first to propose a stealthy CAT that remains hidden within the system’s visual processing pipeline. Once triggered, it disrupts task flow, increases task completion time and cognitive demand, and elevates frustration, ultimately reducing trust in the assistant.

Generalizability of CAT: Our proposed CAT generalizes beyond the four tasks and the device (Magic Leap 2) evaluated in Section 5, as it targets the camera-to-LVLM visual pipeline common to modern XR and mixed-reality cognitive assistants rather than device-specific interfaces or task-specific features. The same attack could therefore affect assistants deployed on platforms such as Microsoft HoloLens, Meta Quest, and Apple Vision Pro. By perturbing a single element of the category-level semantic scene representation, CAT can alter the assistant’s interpretation of task-relevant objects, attributes, or spatial relationships, thereby modifying its guidance across diverse tasks. For instance, in LVLM-guided intelligent XR cognitive-assisted furniture assembly, changing the perceived label or orientation of two similar panels could cause the assistant to recommend the wrong part, leading to misalignments that emerge only at later stages and require time-consuming disassembly and rework. Similarly, in medication sorting, misperceiving a pill’s color (e.g., yellow as white) could direct the user to place the wrong pill in a scheduled time slot, risking missed doses or incorrect dosages. Likewise, in personalized fitness coaching, subtly distorting perceived joint angles could cause a correct posture to be flagged as unsafe, resulting in misleading adjustments that reduce accuracy and compromise workout safety. Across these scenarios, the same stealthy mechanism can increase cognitive load, prolong task completion, degrade performance, and, in safety-critical settings, directly endanger users.

Potential Defenses: To mitigate the CAT identified in this study, we suggest exploring defensive strategies across three layers. At the perception layer, cross-validating multiple XR sensors (e.g., camera, depth, audio, etc.) can expose inconsistencies that reveal tampering, where visual-depth mismatches may indicate forged objects; LVLM analyzers can automatically flag suspicious triggers [63], and content provenance checks alongside hardware-level isolation reinforce input integrity. Building on this, the reasoning layer focuses on bolstering the LVLM’s robustness and transparency through real-time anomaly detection that monitors outputs for logical inconsistencies, explainable AI techniques that provide insight into the LVLM’s decisions to spot anomalies triggered by stealthy attacks [26], and running the LVLM in a trusted execution environment (TEE) to safeguard the reasoning process. Finally, at the interaction layer, a transparent and traceable interface can help users verify guidance by displaying confidence levels or flagging unusual instructions. Since XR visuals can be manipulated to deceive users [57], clearly labeling generative AI-generated content helps users discern real versus artificial cues. Collectively, these layered strategies strengthen cognitive robustness in LVLM-powered intelligent XR systems and enhance system integrity and user trust.

Limitations and Future Work: Our proposed CAT for the LVLM-guided intelligent XR cognitive assistant systems has a few limitations. Firstly, due to time, resource, and budget constraints, we

were unable to conduct a large-scale user study and kept our evaluation limited to two representative tasks. Future work will include a larger participant pool and extend the evaluation to safety-critical tasks (e.g., hazard localization in firefighting, route planning in search and rescue, etc.). Another limitation of our proposed attack is that we evaluated its effectiveness within a fixed window, i.e., a 60-second window, during the user study to analyze participants’ behavior and task performance. In the future, we will evaluate and analyze the impact of CAT across varying time windows. Our future work will explore the attack across varied XR platforms, settings, and broader task categories with more prolonged exposure durations [5, 43, 53], as well as investigate mitigation strategies presented in Section 7 to enhance the resilience of LVLM-guided intelligent XR cognitive assistant systems against such vulnerabilities.

8 Conclusion

In this work, we proposed a stealthy CAT that targets LVLM-guided intelligent XR cognitive assistants by subtly manipulating system perception to degrade user task performance. Triggered only by the attacker’s predefined visual object (e.g., a white cup with a QR code), the Trojan alters visual input to mislead the LVLM’s decision-making, increasing cognitive load and disrupting the immersive flow of the XR experience. We demonstrated the effectiveness of this attack with an LVLM-guided intelligent XR cognitive assistant, namely *CogXR* on the Magic Leap 2 headset, across two task categories: a spatial-reasoning task (Rubik’s Cube solving) and a procedural task (cooking). Our user study revealed that the CAT increases task completion time by an average of 43% and 25.2% for the two tasks respectively, compared to non-attack conditions, leading to higher cognitive load and reduced overall performance. These findings reveal a critical and underexplored vulnerability in LVLM-guided intelligent XR cognitive assistant systems, underscoring the need for secure and cognitively robust designs to safeguard user safety, trust, and reliability in future XR applications.

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).
- [2] Istiak Ahmed, Ripan Kumar Kundu, and Khaza Anuarul Hoque. 2025. Adversarial VR: An Open-Source Testbed for Evaluating Adversarial Robustness of VR Cybersickness Detection and Mitigation. In *2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE, 281–288.
- [3] Abdullah Al Arafat, Zhishan Guo, and Amro Awad. 2021. Vr-spy: A side-channel attack on virtual key-logging in vr headsets. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*. IEEE, 564–572.
- [4] Anduril Industries. 2025. Anduril and Meta Team Up to Transform XR for the American Military. <https://www.anduril.com/news/anduril-and-meta-team-up-to-transform-xr-for-the-american-military>. Accessed: 2025.
- [5] Majid Behravan, Krešimir Matković, and Denis Gračanin. 2025. Generative AI for context-aware 3D object creation using vision-language models in augmented reality. In *2025 IEEE International Conference on AI and eXtended and Virtual Reality (AIxVR)*. IEEE, 73–81.
- [6] Rishika Bhagwatkar, Shravan Nayak, Pouya Bashivan, and Irina Rish. 2024. Improving Adversarial Robustness in Vision-Language Models with Architecture and Prompt Design. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. 17003–17020.
- [7] Swarup Bhunia, Michael S Hsiao, Mainak Banga, and Seetharam Narasimhan. 2014. Hardware Trojan attacks: Threat analysis and countermeasures. *Proc. IEEE* 102, 8 (2014), 1229–1247.
- [8] Elise Bonnal, Wen-jie Tseng, Mark McGill, Eric Lecolinet, Samuel Huron, and Jan Gugenheimer. 2023. Memory manipulations in extended reality. In *Proceedings*

- of the 2023 CHI Conference on Human Factors in Computing Systems. 1–20.
- [9] Andy Bonnetto, Haozhe Qi, Franklin Leong, Matea Tashkowska, Mahdi Rad, Solaiman Shokur, Friedhelm Hummel, Silvestro Micera, Marc Pollefeys, and Alexander Mathis. 2025. EPFL-Smart-Kitchen-30: Densely annotated cooking dataset with 3D kinematics to challenge video and language models. *arXiv preprint arXiv:2506.01608* (2025).
 - [10] Peter Casey, Ibrahim Baggili, and Ananya Yarramreddy. 2021. Immersive Virtual Reality Attacks and the Human Joystick. *IEEE Transactions on Dependable and Secure Computing* 18, 2 (2021), 550–562. doi:10.1109/TDSC.2019.2907942
 - [11] Yasra Chandio, Noman Bashir, and Fatima M Anwar. 2024. Stealthy and practical multi-modal attacks on mixed reality tracking. In *2024 IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality (AIxVR)*. IEEE, 11–20.
 - [12] Kaiming Cheng, Jeffery F Tian, Tadayoshi Kohno, and Franziska Roesner. 2023. Exploring user reactions and mental models towards perceptual manipulation attacks in mixed reality. In *32nd USENIX Security Symposium (USENIX Security 23)*, 911–928.
 - [13] Jan Clusmann, Dyke Ferber, Isabella C Wiest, Carolin V Schneider, Titus J Brinker, Sebastian Foersch, Daniel Truhn, and Jakob Nikolas Kather. 2025. Prompt injection attacks on vision language models in oncology. *Nature Communications* 16, 1 (2025), 1239.
 - [14] Badhan Chandra Das, M Hadi Amini, and Yanzhao Wu. 2025. Security and privacy challenges of large language models: A survey. *Comput. Surveys* 57, 6 (2025), 1–39.
 - [15] Leon Davis and Usman Aslam. 2024. Analyzing consumer expectations and experiences of Augmented Reality (AR) apps in the fashion retail sector. *Journal of Retailing and Consumer Services* 76 (2024), 103577.
 - [16] Defense Advanced Research Projects Agency. 2025. Intrinsic Cognitive Security. <https://www.darpa.mil/research/programs/intrinsic-cognitive-security>. Accessed: 2025.
 - [17] Shane Dirksen, Radha Kumaran, You-Jin Kim, Yilin Wang, and Tobias Höllerer. 2025. Modeling Object Attention in Mobile AR for Intrinsic Cognitive Security. In *Proceedings of the Twenty-sixth International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing*. 479–484.
 - [18] Abhinav Dugar, R Sohanraj, Sparsh Agarwal, Sanket Salvi, and Mrunali Borkar. 2024. Augmented Reality-based Sudoku Solver with Training Module to Improve Cognitive Skills. In *2024 IEEE 3rd World Conference on Applied Intelligence and Computing (AIC)*. IEEE, 60–66.
 - [19] Yalda Ghasemi, Allison Bayro, Justin MacDonald, Heejin Jeong, Joel Reynolds, and Chang S Nam. 2023. Embedding spatial augmented reality in culinary training: A comparative evaluation of sAR kitchen and video tutorials. *IEEE Transactions on Learning Technologies* 17 (2023), 765–775.
 - [20] Kevin Immanuel Gubbi, Banafsheh Saber Latibari, Anirudh Srikanth, Tyler Sheaves, Sayed Arash Beheshti-Shirazi, Sai Manoj PD, Satareh Rafatirad, Avesta Sasan, Houman Homayoun, and Soheil Salehi. 2023. Hardware trojan detection using machine learning: A tutorial. *ACM Transactions on Embedded Computing Systems* 22, 3 (2023), 1–26.
 - [21] Shirin Hajahmadi, Lorenzo Stacchio, Alessandro Giacché, Pasquale Cascarano, and Gustavo Marfia. 2024. Investigating extended reality-powered digital twins for sequential instruction learning: the case of the rubik's cube. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 259–268.
 - [22] Sandra G Hart and Lowell E Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in psychology*. Vol. 52. Elsevier, 139–183.
 - [23] Youcheng Huang, Fengbin Zhu, Jingkun Tang, Pan Zhou, Wenqiang Lei, Jiancheng Lv, and Tat-Seng Chua. 2024. Effective and efficient adversarial detection for vision-language models via a single vector. *arXiv preprint arXiv:2410.22888* (2024).
 - [24] Samuel T King, Joseph A Tucek, Anthony Cozzie, Chris Grier, Weihang Jiang, and Yuanyuan Zhou. 2008. Designing and Implementing Malicious Hardware. *Leet* 8 (2008), 1–8.
 - [25] Ripan Kumar Kundu, Istiak Ahmed, and Khaza Anuarul Hoque. 2025. PrivateXR: Defending Privacy Attacks in Extended Reality Through Explainable AI-Guided Differential Privacy. In *2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 507–517.
 - [26] Ripan Kumar Kundu, Matthew Denton, Genova Mongalo, Prasad Callyam, and Khaza Anuarul Hoque. 2025. Securing Virtual Reality Experiences: Unveiling and Tackling Cybersickness Attacks with Explainable AI. *arXiv preprint arXiv:2503.13419* (2025).
 - [27] Piergiorgio Ladisa, Henrik Plate, Matias Martinez, and Olivier Barais. 2023. SoK: Taxonomy of Attacks on Open-Source Software Supply Chains. In *2023 IEEE Symposium on Security and Privacy (SP)*. IEEE, 1509–1526.
 - [28] Ka Hei Carrie Lau, Sema Sen, Philipp Stark, Efe Bozkir, and Enkelejda Kasneci. 2025. Adaptive Gen-AI Guidance in Virtual Reality: A Multimodal Exploration of Engagement in Neapolitan Pizza-Making. In *Proceedings of the 27th International Conference on Multimodal Interaction*.
 - [29] Siyuan Liang, Jiawei Liang, Tianyu Pang, Chao Du, Aishan Liu, Mingli Zhu, Xiaochun Cao, and Dacheng Tao. 2025. Revisiting Backdoor Attacks against Large Vision-Language Models from Domain Shift. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 9477–9486.
 - [30] Daizong Liu, Mingyu Yang, Xiaoye Qu, Pan Zhou, Yu Cheng, and Wei Hu. 2024. A survey of attacks on large vision-language models: Resources, advances, and future trends. *arXiv preprint arXiv:2407.07403* (2024).
 - [31] Yupei Liu, Yuqi Jia, Runpeng Geng, Jinyuan Jia, and Neil Zhenqiang Gong. 2024. Formalizing and benchmarking prompt injection attacks and defenses. In *33rd USENIX Security Symposium (USENIX Security 24)*. 1831–1847.
 - [32] Weimin Lyu, Lu Pang, Tengfei Ma, Haibin Ling, and Chao Chen. 2024. TrojVlm: Backdoor attack against vision language models. In *European Conference on Computer Vision*. Springer, 467–483.
 - [33] Isaias Majil, Mau-Tsuen Yang, and Sophia Yang. 2022. Augmented reality based interactive cooking guide. *Sensors* 22, 21 (2022), 8290.
 - [34] Ülkü Meteriz-Yıldiran, Necip Fazıl Yıldiran, Amro Awad, and David Mohaisen. 2022. A keylogging inference attack on air-tapping keyboards in virtual environments. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, 765–774.
 - [35] Nicole Modic. 2024. How To Make The Perfect Pan-Seared Steak. <https://kaleju.nkie.com/how-to-make-the-perfect-pan-seared-steak/>. Accessed: 2024-05-15.
 - [36] Blessing Odeleye, George Loukas, Ryan Heartfield, and Fotios Spyridonis. 2021. Detecting framerate-oriented cyber attacks on user experience in virtual reality. In *1st International Workshop on Security for XR and XR for Security*. 1–5.
 - [37] Marc Ohm, Henrik Plate, Arnold Sykosch, and Michael Meier. 2020. Backstabber's knife collection: A review of open source software supply chain attacks. In *Detection of Intrusions and Malware, and Vulnerability Assessment (DIMVA)*. Springer, 23–43.
 - [38] OWASP Gen AI Security Project. 2025. LLM01:2025 Prompt Injection. <https://genai.owasp.org/llm01-prompt-injection/>.
 - [39] Joseph O'Hagan, Jan Gugenheimer, Florian Mathis, Jolie Bonner, Richard Jones, and Mark McGill. 2024. A Viewpoint on the Societal Impact of Everyday Augmented Reality and the Need for Perceptual Human Rights. *IEEE Security & Privacy* 22, 1 (2024), 64–68.
 - [40] Parts Town. 2023. Levels of Steak Doneness. Accessed: 2025-08-01. <https://www.partstown.com/about-us/levels-of-steak-doneness>
 - [41] Lauren L Prill, Lindsey N Drey, Emily A Rice, Brittany A Olson, John M Gonzalez, Jessie L Vipham, Michael S Chao, Phillip Bass, Michael Colle, Travis O'Quinn, et al. 2019. Do published cooking temperatures correspond with consumer and chef perceptions of steak degrees of doneness? *Meat and Muscle Biology* 3, 1 (2019).
 - [42] Mahya Qorbani, Kamran Paynabar, and Mohsen Moghaddam. 2025. Teaching LLMs to See and Guide: Context-Aware Real-Time Assistance in Augmented Reality. [arXiv:2511.00730 \[cs.HC\]](https://arxiv.org/abs/2511.00730) <https://arxiv.org/abs/2511.00730>
 - [43] Zhehan Qu, Ryleigh Byrne, and Maria Gorlatova. 2024. "Looking" into Attention Patterns in Extended Reality: An Eye Tracking-Based Study. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 855–864.
 - [44] RageAgainstThePixel. 2026. com.openai.unity: A Non-Official OpenAI REST Client for Unity (UPM). <https://github.com/RageAgainstThePixel/com.openai.unity>. GitHub repository. Accessed: 2026-03-12.
 - [45] Nornadiah Mohd Razali, Yap Bee Wah, et al. 2011. Power comparisons of shapiro-wilk, kolmogorov-smirnov, lilliefors and anderson-darling tests. *Journal of statistical modeling and analytics* 2, 1 (2011), 21–33.
 - [46] Kristopher W Reese, Taylor Kulp-McDowall, Michael Majurski, Tim Blattner, Derek Juba, Peter Bajcsy, Antonio Cardone, Philippe Dessauw, Alden Dima, Anthony J Kearsley, et al. 2026. Trojans in Artificial Intelligence (TrojAI) Final Report. *arXiv preprint arXiv:2602.07152* (2026).
 - [47] Haocheng Ren, Muzhe Wu, Gregory Thomas Croisdale, Anhong Guo, and Xu Wang. 2025. Rubikon: Intelligent Tutoring for Rubik's Cube Learning Through AR-enabled Physical Task Reconfiguration. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference*. 3549–3562.
 - [48] Maha Sajid, Syed Ibrahim Mustafa Shah Bukhari, Bo Ji, and Brendan David-John. 2025. Just stop doing everything for now!: Understanding security attacks in remote collaborative mixed reality. *arXiv preprint arXiv:2501.16505* (2025).
 - [49] Yangming Shi, John Kang, Pengxiang Xia, Oshin Tyagi, Ranjana K Mehta, and Jing Du. 2021. Spatial knowledge and firefighters' wayfinding performance: A virtual reality search and rescue experiment. *Safety science* 139 (2021), 105231.
 - [50] Carter Slocum, Yicheng Zhang, Nael Abu-Ghazaleh, and Jiasi Chen. 2023. Going through the motions: {AR/VR} keylogging from user head motions. In *32nd USENIX Security Symposium*. 159–174.
 - [51] Carter Slocum, Yicheng Zhang, Erfan Shayegani, Pedram Zaree, Nael Abu-Ghazaleh, and Jiasi Chen. 2024. That Doesn't Go There: Attacks on Shared State in {Multi-User} Augmented Reality Applications. In *33rd USENIX Security Symposium (USENIX Security 24)*. 2761–2778.
 - [52] Enox Software. 2014. OpenCV for Unity. <https://assetstore.unity.com/packages/tools/integration/opencv-for-unity-21088>. Accessed: 2025-04-04.
 - [53] Sruti Srinidhi, Edward Lu, and Anthony Rowe. 2024. XaiR: An XR Platform that Integrates Large Language Models with the Physical World. In *2024 IEEE*

- [54] Zihao Su, Kunlin Cai, Reuben Beeler, Lukas Dresel, Allan Garcia, Ilya Grishchenko, Yuan Tian, Christopher Kruegel, and Giovanni Vigna. 2024. Remote Keylogging Attacks in Multi-user {VR} Applications. In *33rd USENIX Security Symposium (USENIX Security 24)*. 2743–2760.
- [55] SM Taheri and Gholamreza Hesamian. 2013. A generalization of the Wilcoxon signed-rank test and its applications. *Statistical Papers* 54 (2013), 457–470.
- [56] Ali Teymourian, Andrew M Webb, Taha Gharaibeh, Arushi Ghildiyal, and Ibrahim Baggili. 2025. SoK: Come Together—Unifying Security, Information Theory, and Cognition for a Mixed Reality Deception Attack Ontology & Analysis Framework. *arXiv preprint arXiv:2502.09763* (2025).
- [57] Wen-Jie Tseng, Elise Bonnal, Mark McGill, Mohamed Khamis, Eric Lecolinet, Samuel Huron, and Jan Gugenheimer. 2022. The dark side of perceptual manipulations in virtual reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [58] Rithik Vir and Parsa Madinei. 2024. ARChef: An iOS-Based Augmented Reality Cooking Assistant Powered by Multimodal Gemini LLM. *arXiv preprint arXiv:2412.00627* (2024).
- [59] Ao Wang et al. 2025. YOLOE: Real-Time Seeing Anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.
- [60] Feiyang Wang, Xiaomin Yu, and Wangyu Wu. 2025. CubeRobot: Grounding Language in Rubik’s Cube Manipulation via Vision-Language Model. In *Companion Proceedings of the ACM on Web Conference 2025*. 2181–2186.
- [61] Yanming Xiu and Maria Gorlatova. 2025. Detecting visual information manipulation attacks in augmented reality: a multimodal semantic reasoning approach. *arXiv preprint arXiv:2507.20356* (2025).
- [62] Yanming Xiu and Maria Gorlatova. 2025. Vision Language Model-Based Solution for Obstruction Attack in AR: A Meta Quest 3 Implementation. In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, 1638–1639.
- [63] Yanming Xiu, Tim Scargill, and Maria Gorlatova. 2024. LOBSTAR: Language Model-based Obstruction Detection for Augmented Reality. In *2024 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE Computer Society, 335–336.
- [64] Yanming Xiu, Tim Scargill, and Maria Gorlatova. 2025. ViDDAR: Vision Language Model-Based Task-Detrimental Content Detection for Augmented Reality. *IEEE Transactions on Visualization and Computer Graphics* (2025), 1–10. doi:10.1109/TVCG.2025.3549147
- [65] Tianyuan Zhang, Lu Wang, Xinwei Zhang, Yitong Zhang, Boyi Jia, Siyuan Liang, Shengshan Hu, Qiang Fu, Aishan Liu, and Xianglong Liu. 2024. Visual adversarial attack on vision-language models for autonomous driving. *arXiv preprint arXiv:2411.18275* (2024).
- [66] Yicheng Zhang, Zijian Huang, Sophie Chen, Erfan Shayegani, Jiasi Chen, and Nael Abu-Ghazaleh. 2025. Evil Vizier: Vulnerabilities of LLM-Integrated XR Systems. *arXiv preprint arXiv:2509.15213* (2025).
- [67] Yudong Zhang, Ruobing Xie, Yiqing Huang, Jiansheng Chen, Xingwu Sun, Zhanhui Kang, Di Wang, and Yu Wang. 2025. Fighting Fire with Fire (F3): A Training-free and Efficient Visual Adversarial Example Purification Method in LVLMS. *arXiv preprint arXiv:2506.01064* (2025).
- [68] Ada Yi Zhao, Aditya Gunturu, Ellen Yi-Luen Do, and Ryo Suzuki. 2025. Guided Reality: Generating Visually-Enriched AR Task Guidance with LLMs and Vision Models. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*. 1–15.